

Complete Invariants of CTMT Covariance Resolution Geometry

The Moduli Dimension $rq + r + q$, the Necessity of Coupling, and the Insufficiency of Canonical Correlations

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Abstract

This note establishes a purely mathematical pillar of CTMT: the classification of covariance resolution geometries up to admissible isomorphism. Fixing the sector dimensions (r, q) , a geometry reduces in a standard frame to a positive covariance Σ with blocks (Σ_R, Σ_N, C) , and admissible isomorphism is the simultaneous action of $O(r) \times O(q)$. We compute the exact dimension of the moduli space, $rq + r + q$, and give a complete invariant on the generic stratum: the two block spectra together with the coupling expressed in the block eigenbases, modulo a discrete sign/permutation gauge. Two consequences are recorded. First, a *necessity theorem*: the resolved–null coupling C is an irreducible part of the invariant; fixing both block spectra still leaves an rq -dimensional family of pairwise non-isomorphic geometries. Second, and correcting a natural over-reading of the earlier atlas paper, the canonical correlations of the whitened coupling are *not* a complete invariant for $r, q \geq 2$: they omit the relative orientation between the coupling and the block eigenframes, a deficit of $\min(r, q) (\max(r, q) - 1) \geq 2$ real invariants. All statements are elementary consequences of the spectral theorem and compact-group action theory, verified numerically. Like Čencov’s theorem for the Fisher metric, this is a structural result that requires no data and does not expire.

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1 Purpose

CTMT treats the resolved-null coupling $C = C_{RN}$ as a first-class object (Papers II–III). Paper III showed that geometries with differing canonical correlations are non-isomorphic, i.e. that canonical correlations *separate*. It did not claim they are *complete*, and it should not have been read that way. The present note settles the classification exactly: it identifies a complete invariant, computes the moduli dimension, proves that coupling is irreducible, and delimits precisely how far canonical correlations go. This is the kind of pillar that is safest to stand on, because it is a theorem of invariant theory with no empirical content to be contested.

2 The classification problem

Definition 2.1 (Reduced form and admissible action). Fix sector dimensions $r = \dim R$, $q = \dim N$, $m = r + q$. In a g -orthonormal frame adapted to $O = R \oplus N$ the reference metric is $g = I$, the sectors are the coordinate subspaces, and a covariance resolution geometry is a positive definite

$$\Sigma = \begin{pmatrix} \Sigma_R & C \\ C^\top & \Sigma_N \end{pmatrix} \succ 0, \quad \Sigma_R \in \text{Sym}(r), \Sigma_N \in \text{Sym}(q), C \in \mathbb{R}^{r \times q}.$$

Admissible isomorphisms fixing the sectors are the residual gauge group $O(r) \times O(q)$, acting by

$$(A, B) \cdot (\Sigma_R, \Sigma_N, C) = (A \Sigma_R A^\top, B \Sigma_N B^\top, A C B^\top).$$

The moduli space is $\mathcal{M}_{r,q} = \{\Sigma \succ 0\} / (O(r) \times O(q))$.

Remark 2.2 (Why this is the right action). A general admissible morphism is a g -isometry mapping $R \rightarrow R$ (hence $N \rightarrow N$) and pushing forward Σ . In the adapted frame such maps are exactly the block-orthogonal (A, B) , acting as above. Reducing to this frame loses nothing: it fixes the metric and sectors and leaves precisely the covariance to be classified.

3 The moduli dimension

Theorem 3.1 (Moduli dimension). *For generic $\Sigma \succ 0$ the stabilizer in $O(r) \times O(q)$ is finite, the orbit has dimension $\binom{r}{2} + \binom{q}{2}$, and*

$$\dim \mathcal{M}_{r,q} = \frac{m(m+1)}{2} - \binom{r}{2} - \binom{q}{2} = rq + r + q.$$

Proof. The space of covariances $\{\Sigma \succ 0\}$ is open in $\text{Sym}(m)$, of dimension $m(m+1)/2$. The group $O(r) \times O(q)$ has dimension $\binom{r}{2} + \binom{q}{2}$. For generic Σ (distinct eigenvalues within Σ_R and within Σ_N ; see Lemma 3.2) the stabilizer is finite, so the orbit is an immersed submanifold of dimension equal to the group dimension. Hence the generic quotient has dimension $\frac{m(m+1)}{2} - \binom{r}{2} - \binom{q}{2}$. With $m = r + q$,

$$\frac{(r+q)(r+q+1)}{2} - \frac{r(r-1)}{2} - \frac{q(q-1)}{2} = \frac{1}{2}(2rq + 2r + 2q) = rq + r + q. \quad \square$$

Lemma 3.2 (Generic stabilizer is finite). *If Σ_R and Σ_N have distinct eigenvalues, the stabilizer of (Σ_R, Σ_N, C) is contained in $(\pm 1)^r \times (\pm 1)^q$ (diagonal sign changes in the block eigenbases), hence finite.*

Proof. $A \Sigma_R A^\top = \Sigma_R$ means A commutes with Σ_R ; for distinct eigenvalues the commutant in $O(r)$ is the diagonal sign group in the eigenbasis of Σ_R , and likewise for B . The extra condition $A C B^\top = C$ can only shrink this finite set. $\square$

4 A complete invariant

Theorem 4.1 (Complete invariant on the generic stratum). *Suppose Σ_R, Σ_N have distinct eigenvalues. Let $\Sigma_R = U_R \Lambda_R U_R^\top$ and $\Sigma_N = U_N \Lambda_N U_N^\top$ be eigendecompositions and set the eigen-coupling*

$$C_e := U_R^\top C U_N \in \mathbb{R}^{r \times q}.$$

Then a complete invariant of the $O(r) \times O(q)$ -orbit is

$$\tau(\Sigma) = (\text{spec } \Sigma_R, \text{spec } \Sigma_N, [C_e]),$$

where $[C_e]$ is the class of C_e modulo the discrete group of row sign flips $(\pm 1)^r$, column sign flips $(\pm 1)^q$, and the joint permutations that reorder equal-labelled eigenvalues. Two generic geometries are admissibly isomorphic if and only if τ agrees. The invariant has $r + q + rq = \dim \mathcal{M}_{r,q}$ real coordinates, matching Theorem 3.1.

Proof. Invariance: $\text{spec } \Sigma_R, \text{spec } \Sigma_N$ are unchanged by conjugation; under (A, B) the eigenbases transform $U_R \mapsto AU_R P_R$, $U_N \mapsto BU_N P_N$ for signed permutations P_R, P_N (fixing the diagonal Λ), so $C_e = U_R^\top C U_N \mapsto P_R^\top U_R^\top A^\top (ACB^\top) BU_N P_N = P_R^\top C_e P_N$, i.e. $[C_e]$ is invariant.

Completeness: suppose $\tau(\Sigma) = \tau(\Sigma')$. Equal spectra give orthogonal A_0, B_0 with $A_0 \Sigma_R A_0^\top = \Sigma'_R$, $B_0 \Sigma_N B_0^\top = \Sigma'_N$; applying (A_0, B_0) we may assume $\Sigma_R = \Sigma'_R$ and $\Sigma_N = \Sigma'_N$ with common eigenbases. The residual gauge is the finite group of Lemma 3.2, acting on the eigen-coupling by row/column sign flips (and eigenvalue-preserving permutations). Equality of $[C_e]$ means C_e and C'_e lie in one such orbit, so a residual (A_1, B_1) carries C to C' ; then $(A_1 A_0, B_1 B_0)$ is the required isomorphism. Dimension: $r + q$ spectral coordinates plus the rq entries of C_e (the discrete quotient does not reduce dimension), totalling $rq + r + q$. $\square$

Remark 4.2 (Non-generic strata). When Σ_R or Σ_N has repeated eigenvalues the block commutant is a positive-dimensional orthogonal group, the stabilizer is larger, and the complete invariant coarsens accordingly (the eigen-coupling is then taken modulo a continuous group acting on the degenerate eigenspaces). The moduli space is thus stratified; Theorem 4.1 describes the open dense generic stratum.

5 Necessity and irreducibility of the coupling

Corollary 5.1 (Coupling is necessary). *No function of (Σ_R, Σ_N) alone determines the admissible isomorphism class. Fixing both block spectra leaves an rq -dimensional family of pairwise non-isomorphic geometries, parametrised by $[C_e]$. Equivalently, the projection $\mathcal{M}_{r,q} \rightarrow (\text{spec } \Sigma_R, \text{spec } \Sigma_N)$ has rq -dimensional fibres.*

Proof. Immediate from Theorem 4.1: with the spectra fixed, $[C_e]$ ranges over an rq -dimensional set and separates isomorphism classes. $\square$

This is the precise sense in which the resolved-null coupling is not a removable “nuisance” term: it is an irreducible coordinate on the moduli space, carrying rq of its $rq + r + q$ dimensions.

6 Canonical correlations are incomplete for $r, q \geq 2$

Theorem 6.1 (Insufficiency of canonical correlations). *Let $\rho_1, \dots, \rho_{\min(r,q)}$ be the canonical correlations, i.e. the singular values of the whitened coupling $\Sigma_R^{-1/2} C \Sigma_N^{-1/2}$. The invariant $(\text{spec } \Sigma_R, \text{spec } \Sigma_N, \{\rho_i\})$ is complete if and only if $\min(r, q) = 1$. For $r, q \geq 2$ it is strictly coarser than τ , missing*

$$rq - \min(r, q) = \min(r, q)(\max(r, q) - 1) \geq 2$$

real invariants, namely the relative orientation between the coupling's singular frame and the block eigenframes.

Proof. The canonical correlations are $O(r) \times O(q)$ -invariant and number $\min(r, q)$, so $(\text{spec } \Sigma_R, \text{spec } \Sigma_N, \{\rho_i\})$ has $r + q + \min(r, q)$ coordinates. By Theorem 3.1 a complete invariant needs $rq + r + q$; the shortfall is $rq - \min(r, q) = \min(r, q)(\max(r, q) - 1)$, which is 0 iff $\max(r, q) = 1$ (equivalently $\min(r, q) = 1$ given $r, q \geq 1$) and ≥ 2 for $r, q \geq 2$. Concretely, for $r = q = 2$ with fixed Λ_R, Λ_N and fixed ρ , rotating the canonical directions relative to the Σ_R eigenbasis changes the group-invariant $\text{tr}(\Sigma_R C \Sigma_N^{-1} C^\top)$ while preserving both spectra and all ρ_i ; the two geometries are therefore non-isomorphic with identical canonical data. (Verified numerically.) $\square$

Remark 6.2 (Relation to Paper III). Paper III used canonical correlations correctly as a *separating* invariant—distinct ρ implies non-isomorphic. Theorem 6.1 records the converse limitation: equal ρ (with equal spectra) does *not* imply isomorphic once $r, q \geq 2$. The complete invariant is τ of Theorem 4.1; canonical correlations are its orientation-blind shadow. In the scalar-coupling case ($\min(r, q) = 1$, e.g. a one-dimensional resolved or null sector) the two coincide and canonical correlations are complete.

7 Scope and significance

Non-claim 7.1 (Pure structure). This note asserts nothing empirical or physical. It classifies covariance resolution geometries up to admissible isomorphism; the moduli dimension and complete invariant are theorems of linear algebra and compact-group actions.

Remark 7.2 (A Čencov-type role). Čencov's theorem singles out the Fisher metric as the unique Markov-invariant metric. The present result plays the analogous structural role one level down: it identifies the complete invariant of resolution geometry and proves that the resolved–null coupling is an irreducible part of it. Together they say that neither the choice of metric (Čencov) nor the role of coupling (here) is arbitrary. As a pillar this is durable precisely because it can never be contradicted by data: it is a statement about the classification itself.

8 Conclusion

Covariance resolution geometries with fixed sector dimensions form a moduli space of dimension $rq + r + q$; a complete invariant on the generic stratum is the pair of block spectra together with the coupling in the block eigenbases, up to a discrete sign/permutation gauge. The resolved–null coupling occupies rq of those dimensions and is therefore irreducible: no function of the block spectra alone can recover the isomorphism class. Canonical correlations, while genuine invariants that separate distinct classes, are incomplete once both sectors have dimension at least two, missing exactly the relative orientation. This settles the classification that Paper III began, corrects the tempting over-reading that canonical correlations are complete, and provides a permanent, data-independent pillar: the coupling is not decoration but an irreducible coordinate of the theory's fundamental object.

References

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